

TRANSITION DYNAMICS BETWEEN PURSUIT AND FIXATION SUGGEST DIFFERENT SYSTEMS

A. E. LUEBKE* and D. A. ROBINSON

Departments of Biomedical Engineering and Ophthalmology, The Johns Hopkins University,
School of Medicine, Baltimore, MD 21205, U.S.A.

(Received 13 July 1987; in revised form 14 December 1987)

Abstract—The offset of smooth pursuit eye movements is very different from the onset. The onset eye velocity is characterized by overshoot and ringing before settling to steady-state velocity. Yet, at pursuit offset, the eye velocity returns smoothly to zero. One reason for this difference may be that the pursuit system is nonlinear and behaves differently during acceleration and deceleration. After testing four subjects, we found no difference between acceleration and deceleration ringing dynamics, except for decelerations to 0 ± 2 deg/sec, suggesting that around zero velocity, the central nervous system switches off pursuit and adopts fixation. This supports the idea that fixation and pursuit represent different neurological systems.

Eye movements Fixation system Smooth pursuit eye movements Smooth pursuit models

INTRODUCTION

Fixation is said to occur when we look at a stationary target with the head still. When a visual target moves, we track it with an eye movement called smooth pursuit, with a velocity that roughly matches the target velocity; this serves to keep the image on or near the fovea. It is not yet known if fixation and pursuit are created by two different systems or if they are both produced by the same system; that is, is fixation simply pursuit of a target moving at zero velocity?

Recent evidence suggests that fixation and pursuit are not generated by the same system. Robinson *et al.* (1986) performed a detailed analysis of the onset of human smooth pursuit movements to provide a data base for modelling the pursuit system. They used step-ramp target motions of unpredictable directions, velocities and timing. The velocities ranged from 5 to 30 deg/sec. They found that the eye accelerated to peak values between 62 (at 5 deg/sec) and 190 deg/sec² (at 30 deg/sec), reached a peak velocity overshoot and then rang at a frequency of 3.8 Hz and settled to a steady-state velocity commensurate with a steady-state gain (eye velocity/target velocity) of 0.95. The peak

velocity of the overshoot above steady-state velocity was 2.3 deg/sec at 5 deg/sec and rose to 4.7 deg/sec at 30 deg/sec. Figure 1 shows a typical response to similar target motion obtained in this study to illustrate these features. Figure 2A shows the averaged, smoothed onsets of pursuit for the three subjects they studied at 20 deg/sec. Subject 1 had a high acceleration and steady-state gain. Subject 3 had a low acceleration and gain. Subject 2 was intermediate. All three rang at 3.8 Hz at all target velocities. Figure 2B shows the responses averaged feature by feature across all three subjects, for all five target velocities to illustrate the nature of the velocity overshoot and ringing. All three of these subjects were experienced oculomotor physiologists.

This review is given because the subject of this report concerns the presence or absence of such overshoot and ringing. The prior study showed that the eye comes to rest in a manner very different from when it responds to the initial target movement as shown in Fig. 1. When pursuit ceases, eye velocity simply returns smoothly to zero; there is no overshoot or ringing. Any quasi-linear model of the pursuit system (e.g. Robinson *et al.*, 1986) would overshoot and ring just as vigorously at tracking offset as at tracking onset; there would be nothing exceptional about a target velocity of 0 deg/sec. The finding that this was not the case suggested to these authors that when target

*Address for correspondence: A. E. Luebke, 355 Woods/Wilmer, The Johns Hopkins Hospital, Baltimore, MD 21205, U.S.A.

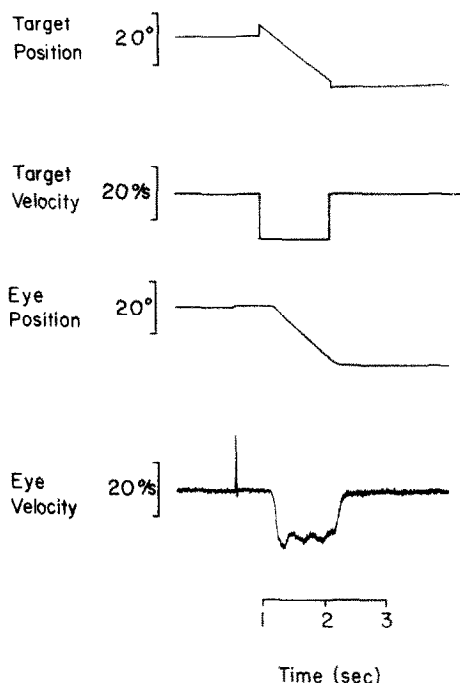


Fig. 1. Eye movements of subject DR resulting from a Rashbass step-ramp of target position. Note that the eye velocity dynamics due to the onset of pursuit (0–20 deg/sec) are very different than the eye velocity dynamics due to the offset of pursuit (20–0 deg/sec).

motion is detected, the pursuit system is turned on and when the target is seen to stop, it is turned off and replaced by fixation. It is possible, however, that a serious, overlooked non-linearity might exist in the pursuit system that causes it to react one way to acceleration and another to deceleration. That hypothesis was tested in this study.

METHODS

Four subjects participated in the experiments; all were experienced in eye-movement recording sessions. The target was a small, bright spot of white light projected via an X-Y mirror galvanometer onto the back of a translucent tangent screen in front of the subject in an otherwise dark room. The head was stabilized with a chin rest. Eye movements were recorded by the magnetic-field/eye-coil method set at a full-scale range where the accuracy was 0.25 deg. The bandwidth was 0–80 Hz (limited by a pen recorder). Horizontal eye velocity was obtained by an analogue differentiator with a bandwidth of 0–50 Hz. The target followed

a step-ramp time course first suggested by Rashbass (1961): it stepped in one direction and immediately moved at a constant velocity in the other. The step amplitude and ramp velocity were adjusted so that the target recrossed its starting position in 0.2 sec, a value intended to allow a pursuit movement to occur without the interruption of a catch-up saccade.

To obtain a long duration for each trial, the target started from an eccentric position and moved centrally. Thus, the subject knew the direction but not the timing or speed of the target motion. The target started unexpectedly, moved at a constant velocity (10, 20 or 30 deg/sec) for a random time interval between 0.4 and 1.0 sec, then switched to a different velocity (between 10 and 30/sec), and then stopped unexpectedly. Velocities were randomized. Pursuit intervals were long enough (greater

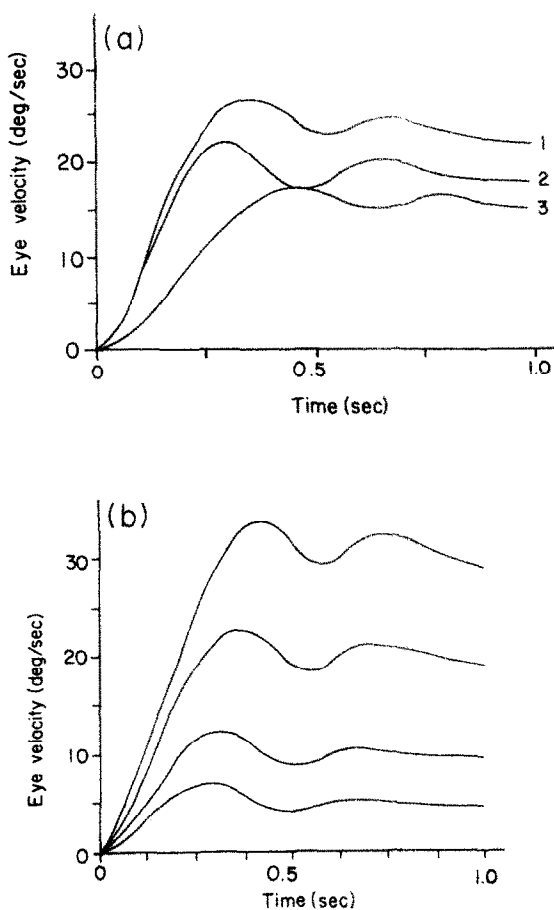


Fig. 2. A. Ramp responses of three subjects for a target velocity of 20 deg/sec to show intersubject similarities and differences. The data have been smoothed by averaging important features. B. Data with smoothed feature points averaged for all three subjects for ramp target velocities of 5, 10, 20 and 30 deg/sec. From Robinson *et al.* (1986).

than 400 msec) for eye velocity to reach steady-state. The step was adjusted whenever the target changed velocity in such a way as to reduce saccades. Target movement was computer-controlled with a PDP 11/73. Eye position, target position, and eye velocity were displayed on a pen recorder and analyzed by hand.

RESULTS

The sample record in Fig. 3 shows most of the features to be mentioned; initially, an increasing rise in eye velocity to a peak acceleration, velocity overshoot and ringing settling to a steady-state value, then a change in eye velocity with further overshoot and ringing due to a step change of target velocity, and finally a smooth decay of velocity to zero with no overshoot and ringing when the target stopped.

The latencies for pursuit initiation were 141 msec with a range of 105–200 msec. The latency that occurred when the target changed from the first to the second velocity was 131 msec with a range of 108–150 msec. These latencies are higher than 100 msec, a value earlier reported for pursuit by Carl and Gellman (1987). A Rashbass step accompanied all pursuit velocity changes so, as suggested by these authors, the step may have caused an apparent increase in the latencies.

All subjects overshoot the steady-state tracking velocity when they initially began to pursue and when the pursuit velocity unexpectedly increased or decreased in the 10–30 deg/sec range. The amount of overshoot due to a velocity step of 20 deg/sec (30–10 or 10–30 deg/sec) was statistically independent, using the Student *t*-test, of whether the step caused an acceleration (10–30 deg/sec) or deceleration (30–10 deg/sec) of eye velocity, with a $P > 0.05^*$ (Table 1). The occurrence of ringing and the amount of overshoot was independent of eye position.

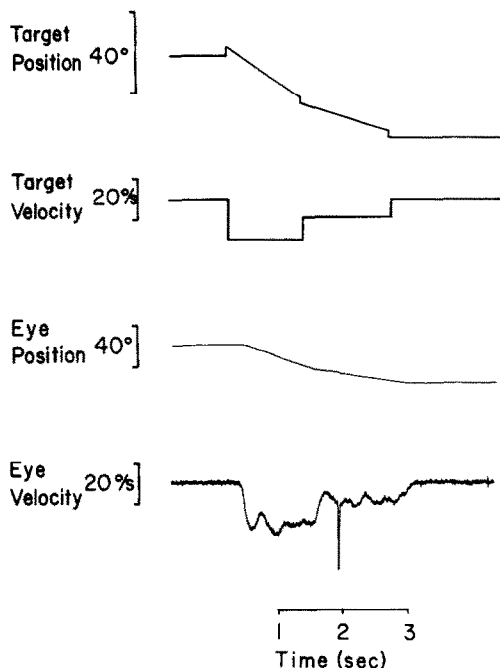


Fig. 3. Eye movements of subject DR resulting from a double-step of target velocity. Note that when the target accelerated from 0–20 deg/sec, the eye velocity dynamics exhibited overshoot and ringing oscillations. When the target decelerated from 20 to 10 deg/sec, similar oscillation dynamics resulted. Yet, when the target decelerated from 10 to 0 deg/sec, no oscillations occurred. Note that Rashbass steps accompanied all velocity changes.

The mean frequency of the oscillations in the eye velocity due to the change in pursuit velocity for each subject for acceleration or deceleration was between 3.26 and 4.23 Hz with an overall mean frequency of 3.9 Hz; similar to the mean frequency of ringing reported earlier by Robinson *et al.* (1986). The oscillation frequency was not different, using the Student *t*-test, due to pursuit accelerations and decelerations (significance level $P > 0.20$).[†] Hence, there is no significant difference between the time course of acceleration and deceleration in smooth pursuit movements, at least between velocities other than near zero.

We wondered how low the target velocity must become before the overshoot and ringing was replaced by the smooth decay seen when the second target velocity was zero. To address this question, we employed a similar two-velocity step-ramp paradigm. In this instance the target velocity was initially 20 deg/sec and then, unexpectedly, the target speed would change to a velocity between 0 and ± 5 deg/sec. The second period of target velocity occurred either in the same direction as the previous

*Using the power test and assuming a Type I error at 0.05 and a Type II error at 0.10, a difference of 0.82 deg/sec is needed for acceleration overshoot and deceleration overshoot to be significantly different. The difference found was 0.13 deg/sec, less than this value, as shown in Table 1, by $\dot{E}_p - \dot{E}_{ss}$.

[†]Using the power test and assuming a Type I error at 0.05 and a Type II error at 0.10, a difference of 0.51 Hz is needed for the acceleration oscillation frequency to differ from the deceleration frequency. The difference found was 0.3 Hz, less than this value, as shown in Table 1.

Table 1. A quantitative analysis of dynamic features occurring in the second pursuit interval. The target moved at a constant velocity of either 10 or 30 deg/sec and then switched unexpectedly to the other velocity. $\dot{E}_p - \dot{E}_{ss}$ is the amount of overshoot, defined as the difference between the velocity of the first overshoot and the steady-state eye velocity. Latencies and frequencies of ringing refer to the response to the second target velocity. Total means refer to the average response due to both acceleration and deceleration of the second target velocity. Group differences refer to the group's response to acceleration versus that of deceleration of the second target velocity. There were six trials at each target motion for each subject

Subject	Target motion (deg/sec)	Overshoot Mean (deg/sec)	Overshoot $\dot{E}_p - \dot{E}_{ss}$ S.D. (deg/sec)	Latency (msec)	Frequency (Hz)
DR	10-30	2.63	1.14	150	4.23
	30-10	3.07	0.49	135	3.26
DZ	10-30	2.08	0.47	148	4.02
	30-10	1.68	0.30	129	3.81
AL	10-30	3.44	0.37	122	3.79
	30-10	2.39	1.04	108	3.69
MF	10-30	2.35	0.14	140	4.17
	30-10	2.86	0.41	120	4.22
Group means	10-30	2.63	0.53	140	4.05
	30-10	2.50	0.56	123	3.75
Total means		2.57	0.55	131.5	3.90
Group differences		0.13			0.30

velocity (i.e. 0.5-5 deg/sec) or it changed direction (-0.5-5 deg/sec).

If overshoot increased with the size of a velocity step, then a step of eye velocity from 20 to 10 deg/sec (step size of 10 deg/sec), as shown in Fig. 3, should exhibit a smaller amplitude of ringing than a step from 20 to 2 deg/sec (step of 18 deg/sec) and a step from 20 to -5 deg/sec (step of 25 deg/sec) should have an even higher amplitude of overshoot and ringing. This did not occur. When the second target velocity approached zero, the number of trials exhibiting overshoot and ringing decreased and, of course, disappeared completely at 0 deg/sec. Figure 4 shows sample records of target velocity steps from 20 to 4, 2, 0, -2 and -4 deg/sec for subject DR. These records were chosen to illustrate the general trend described below. In any individual trial, overshoot and ringing could be full-blown or absent and these records try to illustrate the gradual disappearance of this phenomenon as the second target velocity approached zero in the sense that the amplitude of ringing and overshoot decreased.

Part of this behavior is shown more quantitatively in Table 2. The number of trials displaying any ringing and overshoot of target velocity decreased with the second target velocity, regardless of the direction of that target motion. There was no abrupt change from ringing to not ringing in any of the subjects tested. This transition was clear in all subjects when the second target velocity was 5 deg/sec or larger, compared to 0 deg/sec. All subjects, how-

ever, did not ring vigorously so that for illustrative purposes, the records in Fig. 4 were taken from the subject who exhibited the most ringing.

DISCUSSION

These data suggest that fixation is not pursuit at 0 deg/sec. There is clearly something unique about zero velocity and when the eye decelerates

Table 2. The percentage of trials displaying any ringing and overshoot of eye velocity decreased as the second target velocity decreased, regardless of direction of the second target motion. In all trials, the target velocity was initially 20 deg/sec; then the target speed would change unexpectedly to a velocity between 0 and ± 5 deg/sec. The second period of smooth pursuit occurred either in the same direction as the previous velocity (0.5-5 deg/sec) or it changed direction (-0.5-5 deg/sec). The number of trials at each velocity was about 18

Second target velocity (deg/sec)	Percentage of trials with oscillations (%)
5.0	71.7
4.0	64.6
3.0	67.2
2.0	45.0
1.0	24.3
0.5	10.6
0.0	0.0
-0.5	0.0
-1.0	10.5
-2.0	42.5
-3.0	57.9
-4.0	77.8
-5.0	63.8

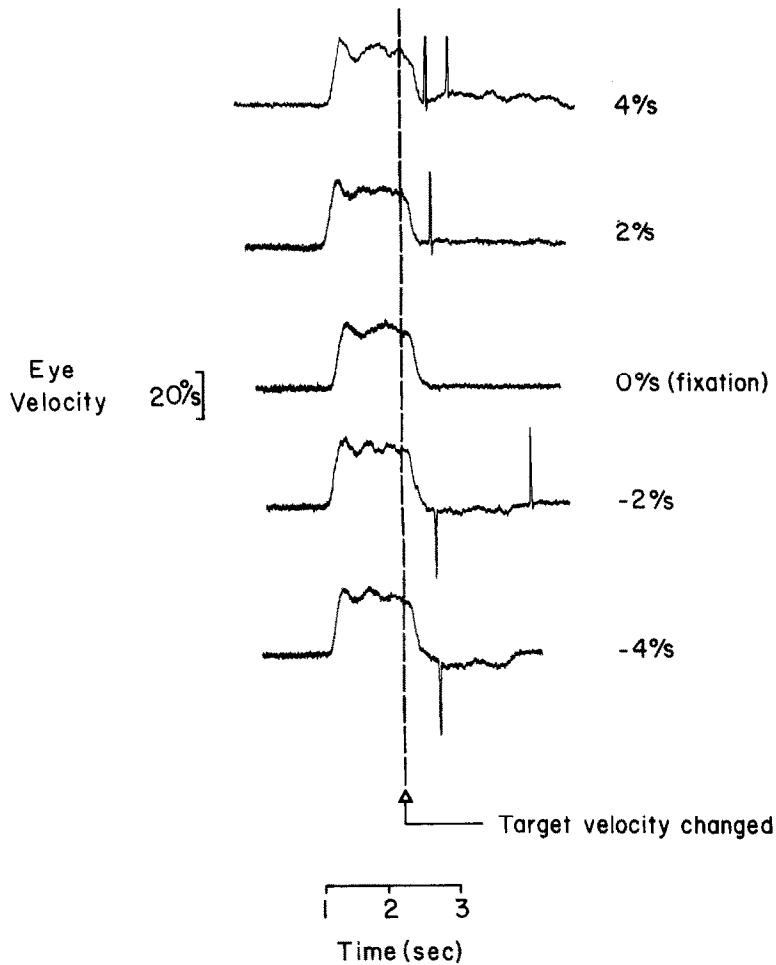


Fig. 4. Eye movements of subject DR resulting from a double-step of target velocity. Initially the target velocity was 20 deg/sec, then the target velocity changed at a randomly chosen time to a velocity between ± 5 deg/sec. Shown here are sample recordings for second velocities of 0, ± 2 and ± 4 deg/sec, all aligned at the time of the second target velocity. The target motion was in the same direction as the initial pursuit for 2 and 4 deg/sec and in the opposite direction for -2 and -4 deg/sec. Note that at velocities from $+2$ to -2 deg/sec, the oscillations seen are very small or nonexistent; not the size expected due to a step change of 18–22 deg/sec.

to this value its dynamics are very different from when it decelerates to other velocities, at least those greater than about ± 2 deg/sec. The data are compatible with the idea that one system, the pursuit system, manages tracking for all velocities and velocity transitions so long as target speed exceeds about 2 deg/sec but as target speed drops below this value and especially when it reaches zero, another system, a fixation system, with different dynamics takes over.

This further suggests that the pursuit system is turned off, to be replaced by fixation, when the target stops and is turned on when the target moves. This is reminiscent of the saccadic sys-

tem which is turned off by pause neurons during fixation and turned on, by inhibition of pause neurons, during a saccade (e.g. Keller, 1977). It is not clear whether the fixation system need be turned off during pursuit or is simply overridden by it. The gray area between ± 2 deg/sec, where ringing and overshoot begin to disappear in a gradual manner, probably reflects the uncertainty of visual-perceptual systems in deciding that the target really did stop. In addition to the overshoot and ringing during pursuit, we were struck by the instability of eye velocity during steady state pursuit. As can be seen in Figs 3 and 4, eye velocity was never really constant during pursuit; little bouts of oscillations at

about 3.9 Hz waxed and waned. At 0 deg/sec the picture was quite different: eye velocity became very steady by comparison. It is well known that slow drifts occur during fixation (Steinman, 1965) with typical values of 0.13 deg/sec, but the oscillations seen during pursuit are on the order of 1.0 deg/sec—an order of magnitude larger. This lends additional support to the idea that fixation and pursuit are mediated by different systems.

There are three other observations of which we are aware that support this idea. (1) Some cells in the dorsolateral pontine nuclei have discharge rates associated with smooth pursuit eye velocity, yet when electrically stimulated they cause the pursuit velocity to increase only if the trained monkey is already pursuing a moving target. If these cells are stimulated when the animal is fixating a target, no eye movement occurs (May *et al.*, 1985). (2) Some cells in the parietal cortex of trained monkeys change their discharge rate during fixation but not pursuit, while others do the opposite (Lynch *et al.*, 1977). The discharge rate did not reflect target or eye metrics, merely the oculomotor state of the animal. (3) Rosenberg *et al.* (1986) found several patients with normal fixation, yet their eyes broke into oscillations similar to those seen in congenital nystagmus whenever they tried to pursue objects moving in a certain direction in a certain field of gaze. This may be an extreme example of patients with congenital nystagmus who have one null zone (a region in which the nystagmus disappears) for fixation but another

when the effort is made to pursue (Del'Osso, 1986). All these findings support the present results that suggest that the pursuit system is switched on and off as needed and that fixation acts in between. Obviously, neurophysiological support is needed to test this idea.

REFERENCES

- Carl J. R. and Gellman R. S. (1987) Human smooth pursuit: stimulus-dependent responses. *J. Neurophysiol.* **57**, 1446–1463.
- Dell'Osso L. F. (1986) Evaluation of smooth pursuit in the presence of congenital nystagmus. *Neuro-ophthal.* **6**, 383–406.
- Keller E. L. (1977) Control of saccadic eye movements by midline brainstem neurons. In *Control of Gaze by Brain Stem Neurons* (Edited by Baker R. and Berthoz A.), pp. 327–337. Elsevier, Amsterdam.
- Lynch J. C., Mountcastle V. B., Talbot W. H. and Yin T. C. T. (1977) Parietal lobe mechanisms for directed visual attention. *J. Neurophysiol.* **40**, 362–389.
- May J. G., Keller E. L. and Crandall W. F. (1985) Changes in eye velocity during smooth pursuit tracking induced by microstimulation in the dorsolateral pontine nucleus of the macaque. *Soc. Neurosci. Abstr.* **11**, 79.
- Rashbass C. (1961) The relationship between saccadic and smooth tracking eye movements. *J. Physiol., Lond.* **159**, 326–338.
- Robinson D. A., Gordon J. L. and Gordon S. E. (1986) A model of the smooth pursuit eye movement system. *Biol. Cybernet.* **55**, 43–57.
- Rosenberg M. L., Kelly B. J., Zee D. S. and Optican L. M. (1986) Unilateral pursuit-induced congenital nystagmus. *Neurology (Suppl. 1)* **36**(4), 251.
- Steinman R. M. (1965) Effects of target size, luminance, and color on monocular fixation. *J. opt. Soc. Am.* **55**, 1158–1165.